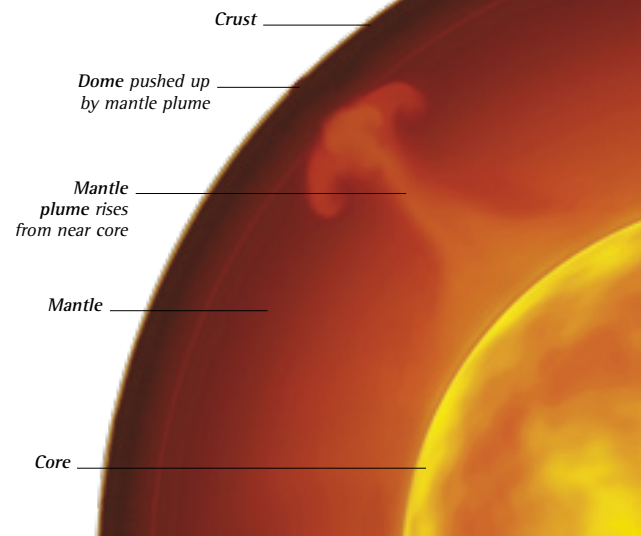


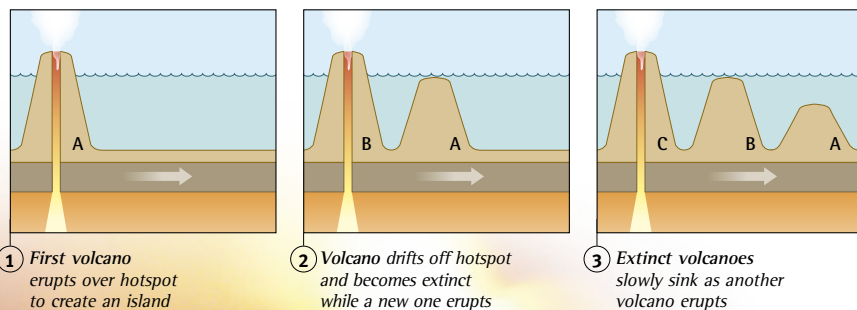
# HOTSPOTS AND SEAMOUNTS

Not all oceanic crust is formed at the spreading rifts of midocean ridges. The hot mantle of the Earth is dotted with extra-hot regions known as mantle plumes. Many of these lie well away from the boundaries between the plates of the Earth's crust. They form stationary hotspots that burn through the crust, erupting molten lava that builds up into volcanoes. As each volcano is carried away from the hotspot by the moving crust it becomes extinct, and another volcano erupts over the hotspot. This process has created chains of volcanic islands and thousands of submerged volcanoes known as seamounts.



## ▲ MANTLE PLUMES

The mantle plumes that cause hotspots probably rise from deep in the mantle near the Earth's core. Each plume pushes up a broad dome in the oceanic crust— 600 miles (1,000 km) across and 1 mile (1.6 km) high in the case of Hawaii—and an even higher volcano erupts from the center of the dome.



## ◀ VOLCANIC CHAINS

The hotspot burns a hole through the Earth's crust, creating a volcano (A). But the mobile crust carries the volcano away from the hotspot, so it becomes extinct and starts sinking. A new one erupts (B), but that is also slowly carried off the hotspot, too. By the time a third volcano appears (C), the oldest may have sunk below the waves.



## ▲ HAWAII

The islands of Hawaii, seen here from space, have been created by the Pacific Ocean plate slipping northwest over a hotspot at about 3 in (9 cm) a year. The hotspot is currently beneath the highly active volcano on the biggest southeastern island. Meanwhile, the extinct volcanoes to the north are slowly sinking, and will become progressively smaller until they dwindle to a long chain of submerged seamounts.

## ◀ FIRE FOUNTAINS

The volcanoes of Hawaii are huge domes that rise from the ocean floor. The largest, Mauna Loa, is higher from base to summit than Mount Everest in the Himalayas. Kilauea, on the flanks of Mauna Loa, is the most active volcano on Earth, constantly erupting fire fountains of fluid basalt lava that flow down to the sea in rivers of molten rock.